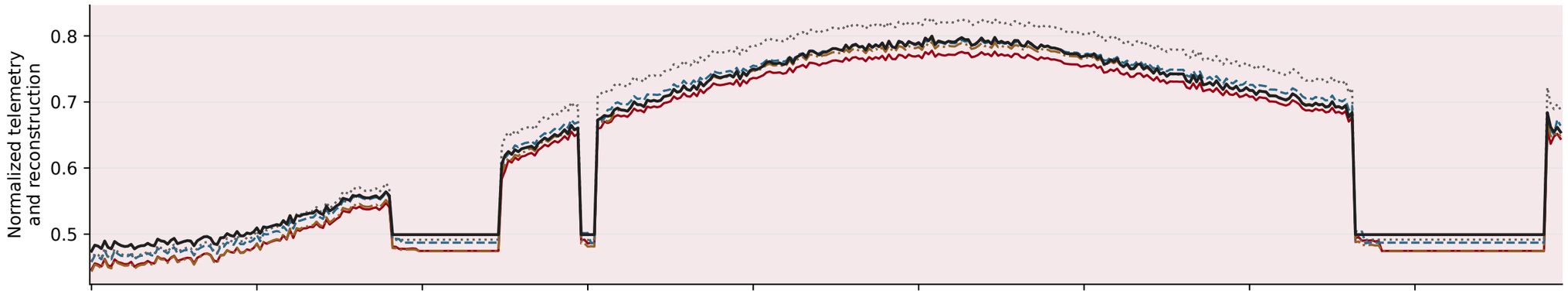
OPS-SAT | 239 | Anomalous fragment

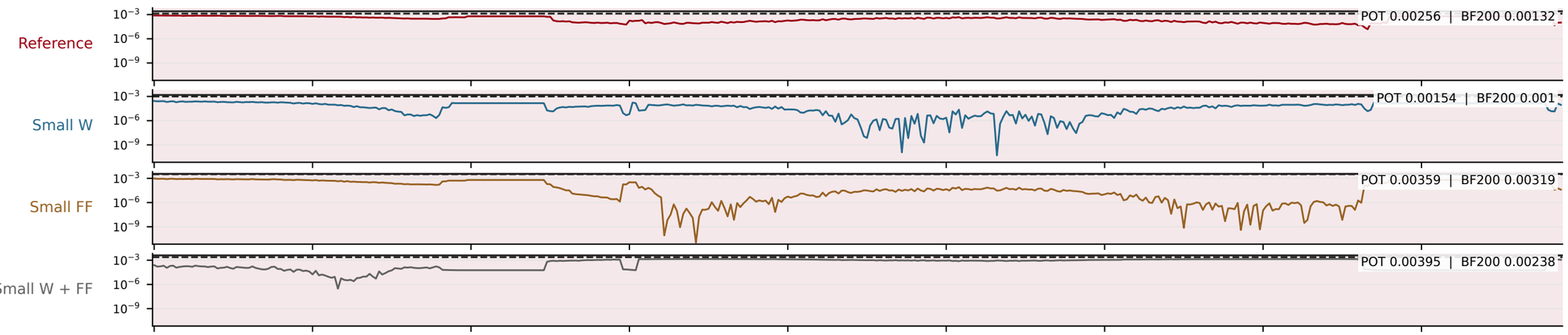
TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5

Observed telemetry — Reference (W10, FF8) — Small W (W5, FF8) — Small FF (W10, FF4) — Small W + FF (W5, FF4)

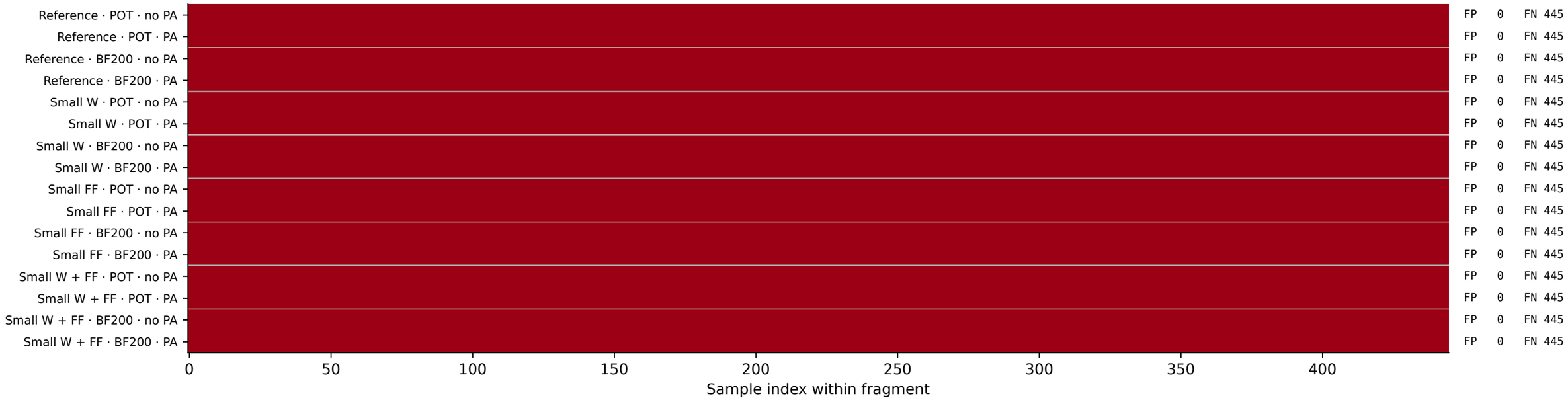
1 Observed telemetry and four reconstructions



2 Anomaly scores and thresholds | solid black: POT; dashed black: BF200



3 False positives and false negatives | same score and threshold before/after PA



False positive: nominal sample flagged — False negative: positive label missed — Correct prediction

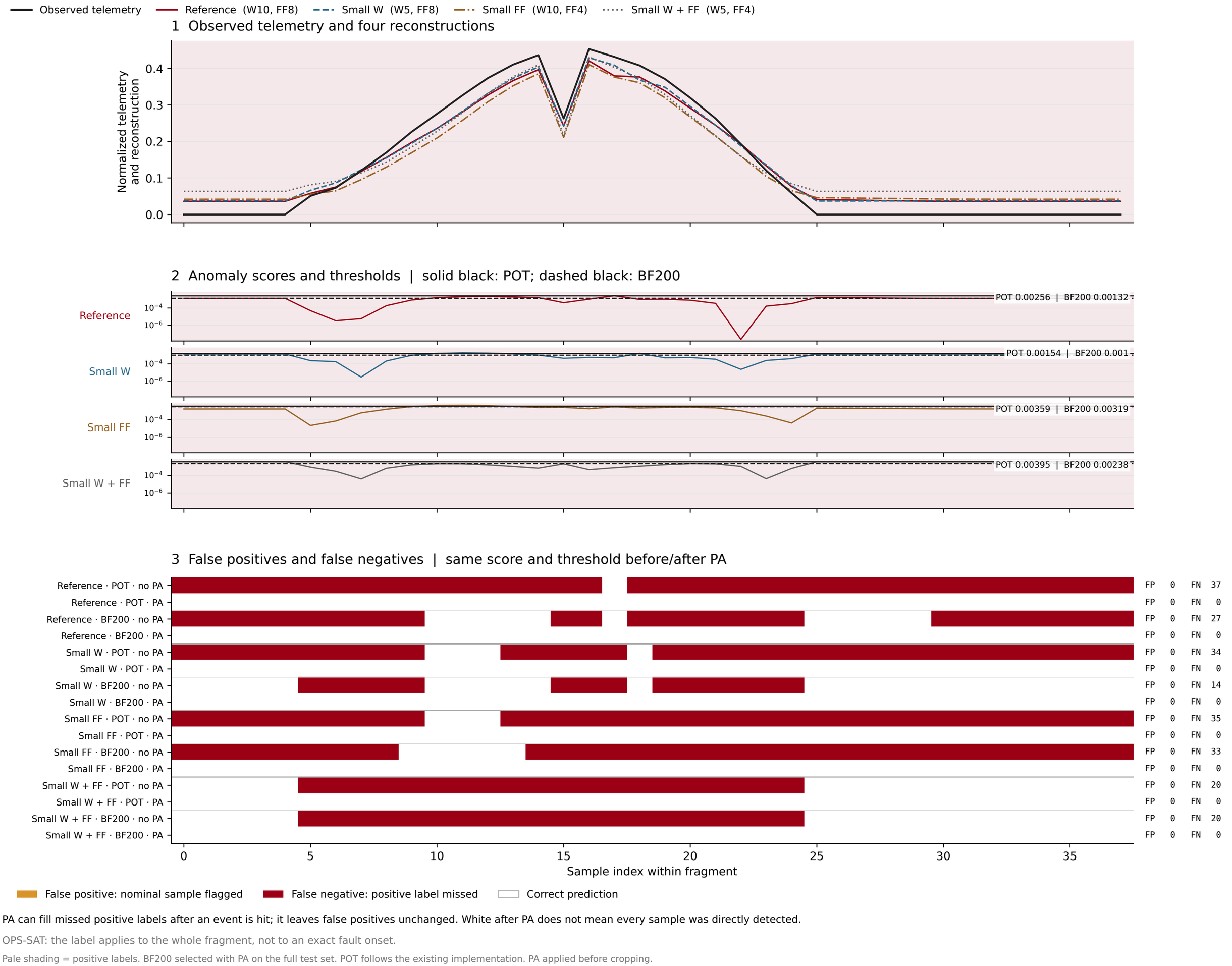
PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

OPS-SAT: the label applies to the whole fragment, not to an exact fault onset.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.

OPS-SAT | 1560 | Anomalous fragment

TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5



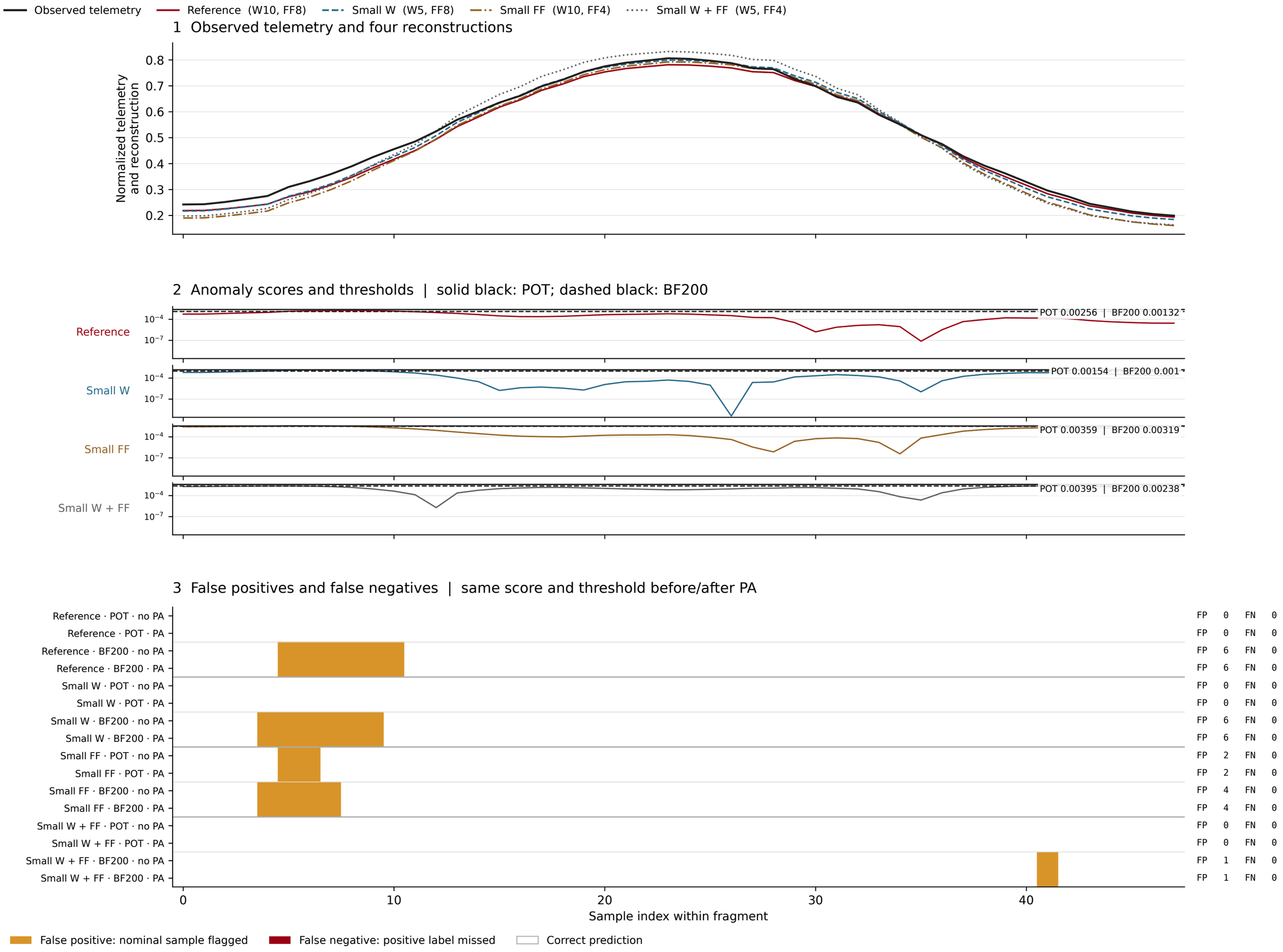
PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

OPS-SAT: the label applies to the whole fragment, not to an exact fault onset.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.

OPS-SAT | 1281 | Nominal fragment

TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5



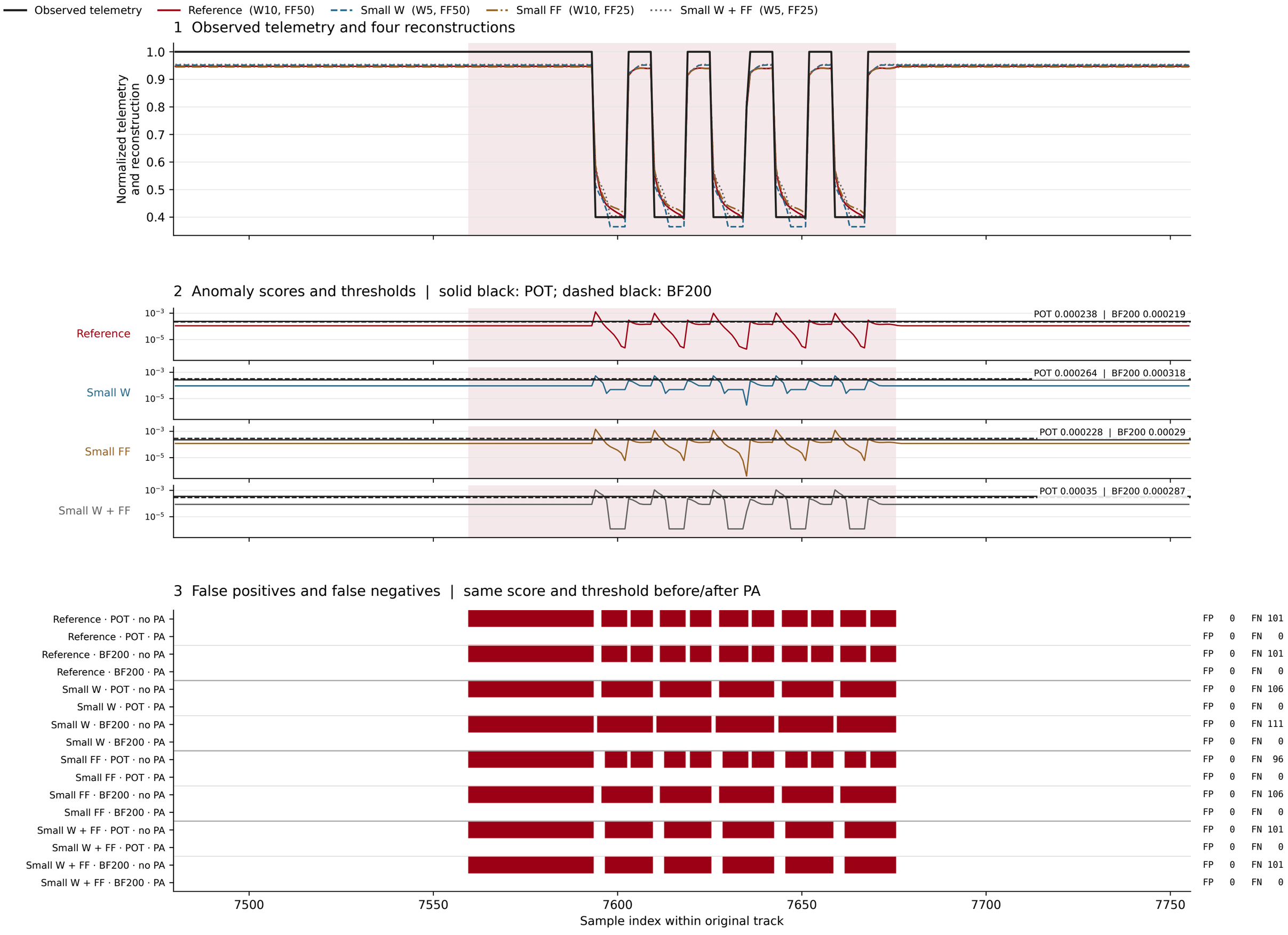
PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

OPS-SAT: the label applies to the whole fragment, not to an exact fault onset.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.

SMAP | G-7 | Labelled anomalous interval

TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5



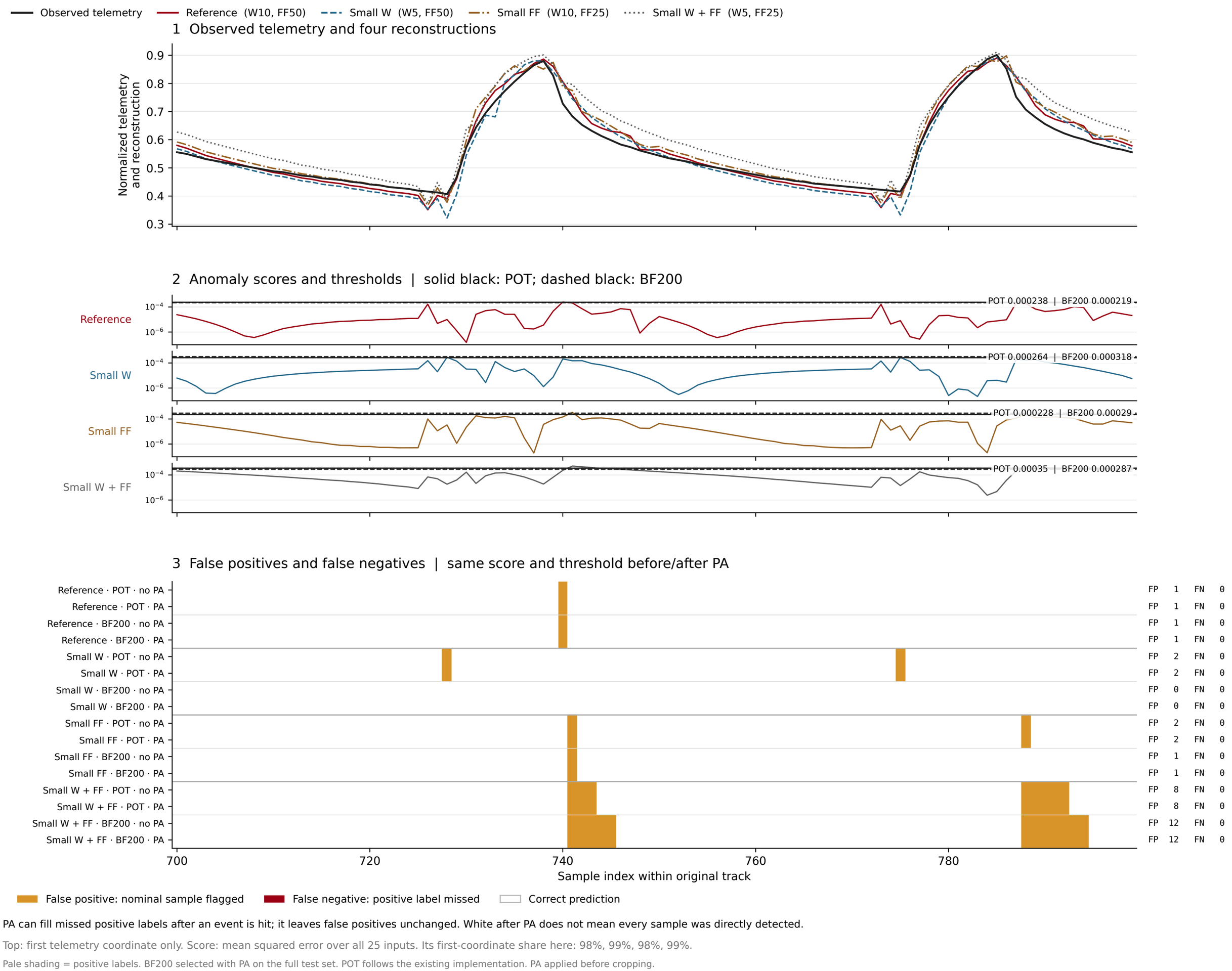
PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

Top: first telemetry coordinate only. Score: mean squared error over all 25 inputs. Its first-coordinate share here: 100%, 100%, 100%, 100%.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.

SMAP | E-4 | Nominal interval

TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5



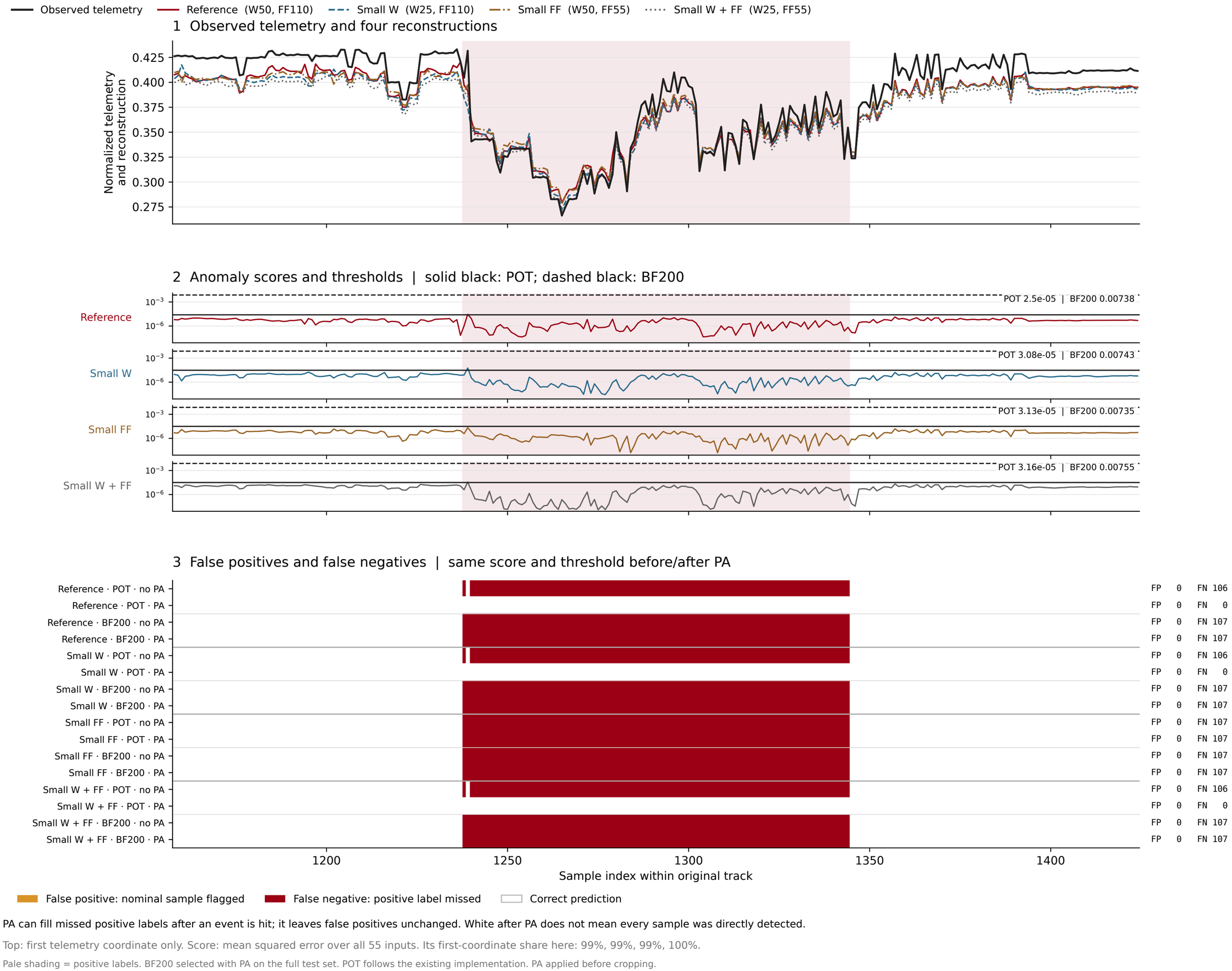
PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

Top: first telemetry coordinate only. Score: mean squared error over all 25 inputs. Its first-coordinate share here: 98%, 99%, 98%, 99%.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.

MSL | P-11 | Labelled anomalous interval

TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5



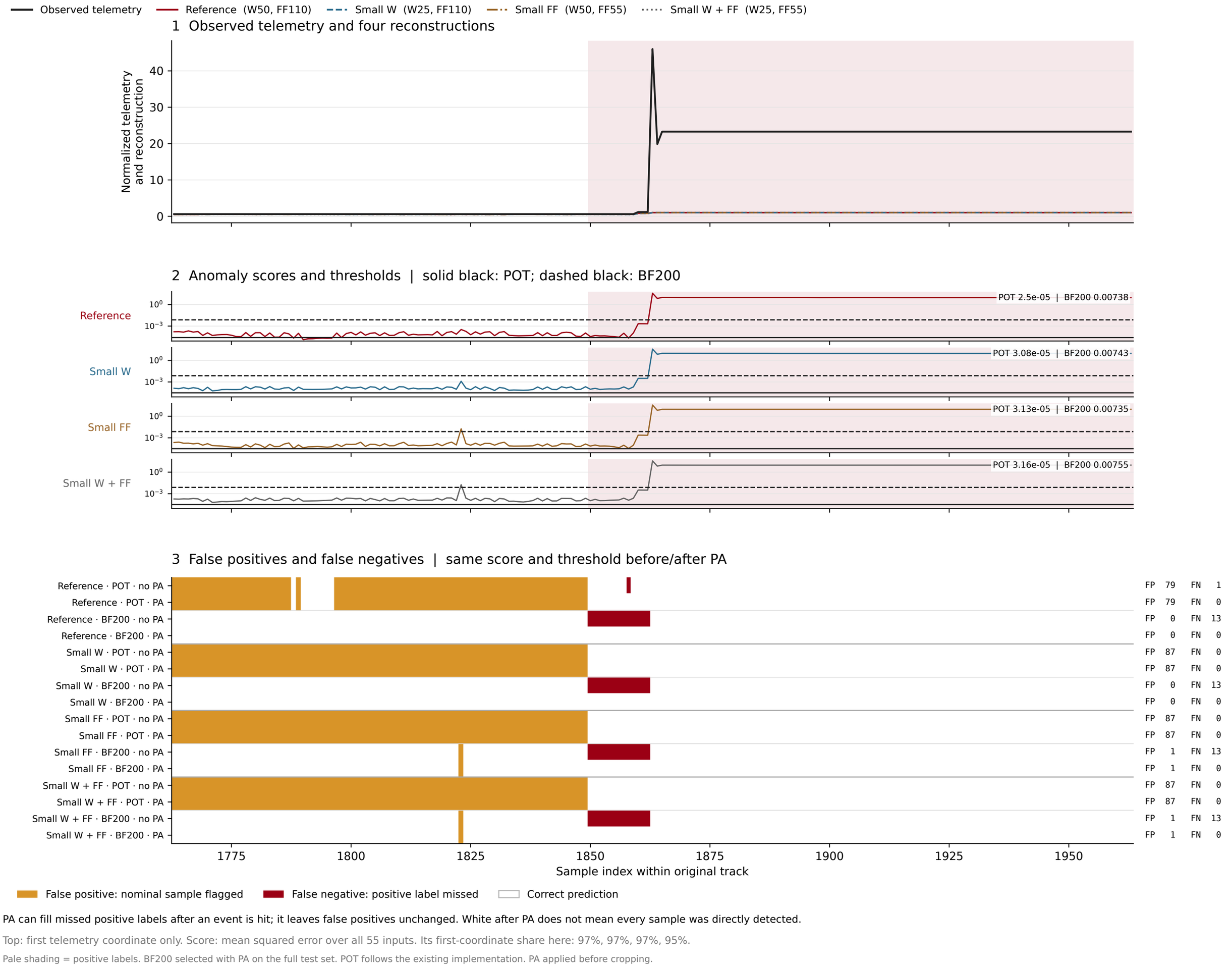
PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

Top: first telemetry coordinate only. Score: mean squared error over all 55 inputs. Its first-coordinate share here: 99%, 99%, 99%, 100%.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.

MSL | M-6 | Extreme-value diagnostic; not a typical case

TranAD+ architecture throughout | seed 2026 | 50 epochs | fixed alpha = 0.5



PA can fill missed positive labels after an event is hit; it leaves false positives unchanged. White after PA does not mean every sample was directly detected.

Top: first telemetry coordinate only. Score: mean squared error over all 55 inputs. Its first-coordinate share here: 97%, 97%, 97%, 95%.

Pale shading = positive labels. BF200 selected with PA on the full test set. POT follows the existing implementation. PA applied before cropping.